Models and Algorithms for EV Charging Station Decision Support

"existing models and algorithms for EV charging station recommendations, demand forecasting, charger utilization optimization, and smart pricing" *Generated by LeapSpace*

EV Charging Decision Support at a Glance

Key Findings Table

Area	Main model families	What the evidence shows	Typical metrics/context	Citations
Station recommendation and siting	Maximal covering, p-median, flow-refueling, bilevel traffic-equilibrium, MILP, GA, NSGA-II, SPEA2, MOEA/D, MCDA	Siting models increasingly combine demand estimation with cost, coverage, congestion, and grid constraints rather than optimizing location alone; contextual inputs such as mobility, traffic, land use, roads, and POIs are common [1] [2] [3] [4] [5] [6] [7].	Coverage, travel cost, detour, waiting, installation cost, voltage/loss impacts; urban, highway, public, and fleet settings [8] [9] [10] [11] [12]	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12]
Demand forecasting	ARIMA, XGBoost, LSTM/GRU/CNN-LSTM, T-GCN, GCN+LSTM, GAT-LSTM-transformer, TFT, Autoformer, LSTM-Transformer	Deep and graph-based models often outperform simple baselines, but not universally; strong temporal baselines such as XGBoost can match or beat more complex models depending on horizon and dataset [13] [14] [15] [16] [17] [18] [19] [20].	RMSE, MAE, MAPE, R2, rolling-origin and train/validation/test splits; hourly to multi-day forecasting [20] [21] [22] [23]	[13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23]
Utilization optimization and queue management	Queueing models, MILP, tabu search, simulated annealing, GA, PSO, greedy best-first search, RL	Queue-aware assignment and scheduling can reduce peak load, improve waiting-time prediction, and increase delivered energy or utilization; finite-population and impatience-aware queue models add realism missing from simpler FCFS/infinite-queue assumptions [24] [25] [26] [27] [28] [29] [30] [31].	Waiting time, queue length, served energy, occupancy, utilization, abandonment, fairness, load variance [29] [32] [33] [34] [35]	[24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35]
Smart pricing and demand response	TOU, real-time, congestion-based, stochastic, bilevel, rolling-horizon, RL pricing	Pricing can shift load, reduce congestion, and improve revenue, but elasticity is context-dependent; RL and adaptive pricing often outperform static tariffs in simulation [36] [37] [38] [39] [40] [41].	Peak reduction, revenue, load variance, waiting, customer choice, congestion share [37] [41] [42] [43] [44]	[36] [37] [38] [39] [40] [41] [42] [43] [44]
Integrated frameworks	Forecasting + pricing + charging/discharging control + V1G/V2G + DER/storage optimization	The field is converging toward end-to-end pipelines where forecasts feed scheduling, assignment, and tariff decisions; these integrated systems report gains in peak reduction, renewable utilization, cost, and revenue [1] [36] [41] [45] [46].	Peak/load variance, user cost, renewable share, grid reliability, voltage stability, revenue [45] [46] [47] [48]	[1] [36] [41] [45] [46] [47] [48]

Direct Answer

Existing EV charging decision-support algorithms fall into four main groups. First, **station recommendation and siting** uses coverage/location-allocation models, flow-refueling, bilevel traffic models, MILP, and evolutionary multi-objective methods, often informed by mobility, traffic, land-use, and POI data [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12]. Second, **demand forecasting** uses classical time-series and ML baselines alongside LSTM/GRU/CNN-LSTM, graph neural models, TFT, Autoformer, and hybrid Transformer systems [13] [14] [15] [23] [49] [50] [51] [52]. Third, **charger utilization optimization** relies on queueing, scheduling, routing, metaheuristics, mathematical programming, and increasingly RL to reduce waiting and improve throughput [25] [26] [27] [28] [29] [53] [54]. Fourth, **smart pricing** uses TOU, congestion pricing, stochastic/bilevel optimization, rolling-horizon control, and RL-based tariff adaptation [37] [40] [41] [55] [56] [57].

The strongest cross-cutting pattern is **integration**: forecasting increasingly feeds siting, queue-aware assignment, and dynamic pricing instead of being treated as a separate task [1] [36] [41] [46]. The best practical takeaway is therefore not "pick one best algorithm," but **build a layered system**: strong temporal forecasting baseline first, then add spatial/queueing/pricing intelligence where real spillovers, congestion, or grid constraints justify the complexity [14] [20] [25] [41].
Confidence level: High - supported by multiple direct findings and cross-theme synthesis [1] [13] [24] [36] [46].

Study Scope

This report covers evidence spanning urban, public, highway/intercity, freight, fleet, and smart-grid charging contexts, with methods including forecasting, location-allocation, queueing, MILP, metaheuristics, bilevel optimization, stochastic programming, and reinforcement learning [13] [30] [55] [58] [59] [60]. The literature represented here ranges from foundational EV siting/control formulations to studies reported through 2025 [9] [10] [11] [30] [61].
Confidence level: High - scope is directly stated in the retrieved findings [9] [10] [61].

Assumptions & Limitations

The evidence does **not** include a single standardized benchmark comparing all major siting, forecasting, utilization, and pricing algorithms on one common real-world dataset with shared operational metrics [17] [19] [20]. Many studies also evaluate isolated problem layers on simulation, proprietary data, or toy networks, limiting direct comparability and deployment generalization [62] [63] [64] [65] [66]. In forecasting and pricing specifically, model superiority is context-dependent rather than universal [14] [18] [39] [67].
Confidence level: High - repeatedly stated across review and benchmarking evidence [17] [19] [20] [66].

Suggested Further Research

- Build **end-to-end public benchmarks** linking forecasting, siting, queueing, and pricing under common evaluation protocols [19] [20] [66].

- Test **queue-aware siting and pricing** with finite-population, abandonment, and cross-station spillover realism [30] [31] [34] [58].
- Run **field or randomized deployments** for dynamic pricing, reservation, and RL control beyond synthetic simulation [54] [66] [68].
- Compare **simple temporal baselines vs graph/deep models** under leakage-free rolling-origin validation across cities and horizons [14] [17] [20].
- Study **heterogeneous elasticity and fairness** by station type, trip purpose, SOC, and accessibility conditions [35] [44].

Foundational Context

EV charging decision support is not one problem but a linked stack of decisions: **where to place chargers, how much demand to expect, how to assign and schedule vehicles once they arrive, and how to influence behavior using prices** [1] [36] [69]. Most mature studies therefore mix **prediction** and **optimization** rather than treating them independently [1] [41] [46].

Four methodological building blocks recur throughout the literature:

1. **Spatial demand estimation** from mobility, traffic, land use, POIs, roads, and demographics for siting and recommendation [2] [3] [4] [5] [6] [7].
2. **Temporal and spatiotemporal forecasting** to predict load at charger, station, or regional level [13] [14] [70].
3. **Operational optimization** using queueing, routing, scheduling, and assignment to control congestion and utilization [25] [28] [29].
4. **Pricing and demand response** to shift charging in time or space while respecting user utility and grid constraints [36] [43] [55].

Taken together, these findings suggest a shift from single-task models toward **closed-loop control architectures** in which prediction informs action, and actions reshape the future demand pattern that must again be predicted [1] [41] [46] [49].
Confidence level: High - directly supported by meta-synthesis and multiple application studies [1] [36] [46].

1) Station Recommendation and Siting Models

1.1 Data-driven siting inputs and spatial demand signals

A common pattern in siting studies is to estimate likely charging demand using **commuting patterns, travel chains, traffic patterns, population density, major roads, and POIs** [2] [3] [4] [5] [6] [7]. These inputs are used because charging demand clusters around activity systems, not just around static residential density [2] [3] [6].

One Beijing study exemplifies the newer integrated direction: it combined **charging-demand forecasting, land-use information, graph structure, and multi-objective expansion planning**, reporting that graph-augmented boosting worked best for forecasting and NSGA-II for expansion planning [1]. However, the evidence also notes that no retrieved paper combines **mobility traces, traffic flow, land use, and POIs** in one unified multi-objective siting model with direct comparison against gravity-model or coverage-only baselines [71] [72] [73] [74].

This fragmentation matters: reviews repeatedly report that forecasting, siting, sizing, and grid impact analysis are often handled separately, reducing realism and comparability [19] [62] [63] [64].
Confidence level: High - evidence is direct and consistent [1] [2] [3] [4] [5] [6] [7].

1.2 Optimization formulations for urban and public station placement

Foundational siting work established several durable formulations: **maximal covering**, **flow-refueling location models**, and later capacity/range-uncertainty extensions [10] [11] [75] [76]. Later public-charging models extended these ideas using **bilevel traffic-equilibrium formulations** that jointly model station placement and driver route/recharge behavior under congestion and queueing [12] [77] [78].

Multi-objective evolutionary methods are especially prominent when planners must balance **coverage, service quality, installation cost, maintenance cost, and grid effects** [8] [9]. In Málaga, NSGA-II and SPEA2 delivered competitive service-cost trade-offs in mobility-based planning, though the retrieved evidence does not report demand-capture accuracy versus gravity or coverage-only baselines [71]. GA-based models also explicitly trade off service coverage and cost under budget and capacity constraints [8].

An important extension is **grid-aware siting**. Forecast-informed evolutionary placement models report objectives such as minimizing **active/reactive power losses** and **voltage deviation**, not just travel or access metrics [9] [79]. This indicates that station recommendation is increasingly an infrastructure-and-power co-design problem rather than a pure transport location problem [9] [77].
Confidence level: High - multiple direct formulations and reported trade-offs [8] [9] [10] [11] [12].

1.3 Highway, intercity, and freight corridor deployment

Highway and corridor charging introduces route feasibility, detours, and queueing in ways urban models do not [80] [81]. Empirical studies therefore use **expanded-network**, **metanetwork**, **route-constrained**, and **flow-capturing** formulations [58] [82].

For intercity travel, one expanded-network/variational-inequality model used an approximated queueing-time function and reported **system cost** and an optimality gap below 5% on a toy network and a Yangtze River Delta highway network [58]. Another study contrasted a two-stage model for uncongested highway networks with a network-based bilevel model for congested ones, comparing solution efficiency and routing behavior across two real networks [82].

For long-haul freight, GIS coverage plus location-allocation analysis reported that **27% of identified truck fast-charging stations could cover more than 90% of total highway coverage** [83]. Across highway/intercity studies, the main reported metrics are **coverage, detour/travel time, queueing delay, and system cost**, while direct operator-side throughput/utilization evidence is comparatively sparse [58] [59] [81].
Confidence level: Medium-High - corridor models are clear, but utilization evidence is thinner [58] [81] [83].

1.4 Fleet-oriented and demand-integrated planning

Fleet planning broadens siting from public-accessibility logic to operational continuity under stochastic use [59] [84]. One e-taxi itinerary-interception model explicitly included **stochastic charging demand and queueing delays**, optimizing travel time/cost and recharge delays while positioning itself against node-based access-distance and flow-interception models [59]. Multi-period MILP work also combines **node-based and flow-based demand** to support both local and long-distance travel [84].

Taken together, these findings suggest that fleet contexts require planning models that integrate **where vehicles move, when they need energy, and how queueing affects service continuity** [59] [84].
Confidence level: Medium - directly supported, but fewer studies are surfaced than for public siting [59] [84].

2) Charging Demand Forecasting

2.1 Short-term forecasting with temporal and tree-based models

Short-term forecasting is dominated by comparisons among **ARIMA-style baselines, tree ensembles, and recurrent/deep temporal models** [13] [14] [15]. On a Beijing GPS-derived dataset, LSTM outperformed ARIMA and MLP [15]. But deployment-oriented benchmarking on hourly ZIP-code data found that **temporal signals dominated**, with **XGBoost achieving the lowest RMSE**, while GCN+LSTM and LSTM were close behind [14] [20].

This is one of the clearest conflicts in the evidence: deep models often win, but not always [16] [17] [18] [19]. A practical implication is that charger-demand forecasting should start with a strong, well-tuned non-deep baseline rather than assuming sequence models will dominate [14] [20].
Confidence level: High - direct comparative evidence plus reproducibility studies [14] [15] [16] [17] [20].

2.2 Spatiotemporal and graph neural forecasting

Spatiotemporal models add **region adjacency, demand graphs, POI-based clustering, traffic, weather, and holidays** to capture cross-region interactions [50] [51] [85] [86]. A community-level T-GCN plus traffic/temperature/network-dynamics framework reported improvements of up to **11.78% in MAE**, **19.5% in MAPE**, **11.02% in TDPD**, and **13.27% in R2** [50]. Other graph-attention or hybrid transformer systems also outperform broad baseline sets, though the retrieved excerpts often omit exact uplifts over temporal-only baselines [51] [86].

However, spatial context is not universally decisive. In ZIP-code benchmarking, spatial modeling did not consistently improve over temporal-only methods; GCN+LSTM was competitive but did not clearly dominate LSTM or XGBoost [20]. This supports a conditional conclusion: spatial layers help most when **inter-station spillovers and network effects are strong and well measured** [20] [50].
Confidence level: High - direct evidence includes both positive and conflicting cases [20] [50].

2.3 Long-horizon and transformer-based forecasting

Long-horizon EV charging forecasting is less mature but increasingly uses **Transformer-family models** [49] [52] [87]. On the Boulder public-charger dataset, an Autoformer-based model improved RMSE over LSTM by about **15% at 30 days** and by more than **50% at 90 days** [49]. On ACN public charging data, a hybrid **LSTM-Transformer** beat standalone LSTM and Transformer at **30, 120, and 240 days**, with the largest gain at 30 days [52].

For shorter horizons, TFT on the Palo Alto dataset beat LSTM, CNN, XGBoost, SARIMA, persistence, and seasonal-naive baselines at **1-24 h and 1-7 day** horizons [23]. Taken together, these findings suggest Transformer-family methods are especially promising when long-horizon structure or multiscale seasonality matters [49] [52].
Confidence level: Medium-High - clear positive evidence, but the long-horizon evidence base is still relatively sparse [49] [52] [87].

2.4 Benchmarks, reproducibility, and feature ablations

The retrieved literature includes several public or open-access resources, including **Boulder ZIP-code hourly charging sessions**, **UrbanEV from Shenzhen**, a **public charging-transaction dataset**, and an **open real-world EV dataset with charging-session and geographic records** [20] [88] [89].

One reproducible study is especially important methodologically: it used **leakage-free preprocessing** and **rolling-origin evaluation** to compare persistence, ARIMA, XGBoost, LSTM, GCN, and GCN+LSTM [20]. In that setup, XGBoost had the best RMSE, while spatial GCN had the best MAE [20]. Calendar covariates consistently improved results, while history windows beyond **24 h** added limited benefit [20] [90] [91] [92].

This is a major practical lesson: evaluation design and feature engineering can change which model appears “best” [17] [20].
Confidence level: High - strong direct benchmarking evidence [20].

3) Charger Utilization Optimization and Queue Management

3.1 Queueing models and waiting-time prediction

Queueing realism is increasingly recognized as central to operational performance [25] [30] [31]. A 2025 study found **finite-population queueing** improved waiting-time prediction accuracy, especially during peak demand, relative to infinite-population formulations [30]. Other work models **balking, reneging, and retry behavior** to better represent impatient users and power-capacity constraints [31].

This matters because FCFS/infinite-queue assumptions can miss abandonment and station-choice behavior that directly affect utilization, pricing, and profitability [31] [34].
Confidence level: Medium-High - concept is well supported, but quantified gains beyond waiting prediction are limited in the excerpts [30] [31] [34].

3.2 Scheduling and load-balancing algorithms

Coordinated charging schedules can materially reshape load and service efficiency. One study reported a **33% reduction in uncoordinated charging load** and an **87.5% increase in off-peak load** [24]. Another heuristic scheduler reduced peak load by up to **39.5%** versus Earliest allocation and **10.4%** versus pure Bose allocation [27].

Among optimization methods, **simulated annealing** beat BPAP and Qin-Modified benchmarks in BEB charging scenarios [32]. A refined MILP-based scheduling study with CSA statistically outperformed GA and GWO and improved average delivered energy by up to **89.45%** over recent state-of-the-art methods [29]. GA-based rescheduling reduced cost, charging time, and load variance, including a **33.3%** load-variance reduction versus equal allocation [33]. Hybrid tabu search plus exact programming also significantly outperformed conventional methods [26].

The consistent pattern is that utilization optimization is **multi-objective and constraint-driven**, with no single dominant solver across all cases [26] [29].
Confidence level: High - multiple comparative studies report quantified gains [24] [27] [29] [32] [33].

3.3 Routing-aware charger assignment

Routing and station assignment interact strongly when congestion is endogenous [93] [94] [95]. Studies using queue-aware station choice with **M/M/1 or M/M/c waiting estimates** aim to minimize waiting and congestion rather than only travel distance [94] [96] [97]. A greedy best-first search framework was designed to form more reasonable EV queues, alleviate traffic congestion, and significantly increase station utilization [28].

Simulation evidence also shows that road awareness matters: one multi-agent study found that real road information significantly reduced congestion versus always taking the shortest route [98].
Confidence level: Medium - routing benefits are consistent, but reported metrics are heterogeneous [28] [95] [98].

3.4 Reservation, fairness, and service-level policies

Recent work adds **reservation, fairness, and RL dispatch** to utilization control [35] [53] [54] [99]. A real-world reservation service using time-aware reservations, price forecasting, and curtailment found that DC fast-charging users shifted sessions into lower-cost periods while user participation remained stable [54]. RL dispatch for mobile charging stations improved cumulative reward relative to greedy, queue-based, static, and random baselines in synthetic and realistic cases [53].

Service-level effects are increasingly quantified with **occupancy, server utilization, abandonment, throughput, QoS, and fairness scores** [34] [35]. This is important because high utilization without fairness or QoS can still produce poor user outcomes [35].
Confidence level: Medium-High - strong direction of travel, though much evidence remains simulation-based [35] [53] [54].

4) Dynamic Pricing and Demand Response

4.1 Time-of-use and congestion-based tariffs

TOU tariffs can shift load, but the effect size varies sharply by setting. One study found TOU moved **50-90%** of EV load from peak to off-peak hours, while another found only a **5%** peak reduction when charging start times were randomized within the off-peak window [38] [100]. A stacked grid tariff experiment cut the share of time with grid congestion by **21%** [37].

User acceptance depends not just on price but also on inconvenience. A conjoint study found **price** had the strongest effect on utility, and users were willing to pay **2.4 €-ct/kWh more** to reduce waiting by 10 minutes [42]. This implies pricing effectiveness is mediated by queue conditions and station convenience, not price alone [39] [42].
Confidence level: High - direct evidence includes both strong and weak TOU effects [38] [39] [42] [100].

4.2 Stochastic, bilevel, and rolling-horizon pricing

When pricing must respect uncertain demand, renewables, or market conditions, studies commonly use **stochastic programming, bilevel models, and rolling horizons** [55] [56] [57] [101]. A scenario-based multi-objective stochastic model solved with NSGA-II cut charging cost and peak load while increasing aggregator profit [55]. Rolling-horizon stochastic bidding is used to maximize station profit under price uncertainty while keeping the problem tractable [56].

Bilevel fast-charging pricing is used to balance operator profit against user charging requirements [57]. Notably, one uncertainty-inclusive demand-response formulation smoothed load and lowered user costs but could also raise supplier revenue alongside higher GHG emissions, illustrating that “better pricing” may improve one objective while worsening another [102].
Confidence level: High - the multi-objective trade-off structure is explicit [55] [56] [57] [102].

4.3 Learning-based pricing and adaptive control

Adaptive pricing is one of the clearest growth areas. RL-based frameworks report higher revenue, better load balancing, or lower cost than static or benchmark pricing/scheduling strategies [40] [41] [103] [104]. One DRL framework with demand forecasting increased total daily revenue by **133%** and reduced load variance by **72.37%** versus flat, TOU, and demand-based pricing [41]. Another online model-free RL controller achieved **138.5%** higher charging-station profit than benchmark methods [40].

But there is also cautionary evidence: an agent-based study found simplistic TOU adoption can worsen local substation peaks [105]. This implies that adaptive control outperforms static tariffs partly because it reacts to network conditions that static TOU ignores [41] [105].
Confidence level: High - strong simulated gains, though mostly not from randomized real-world trials [40] [41] [105].

4.4 Field evidence and customer response

Empirical user-response evidence exists but is methodologically mixed. Historical session data show that changing fee structures can alter behavior: a flat-fee-to-payment change increased low-SOC charging, free stations were used about **four times** more than paid ones, and TOU pricing shifted demand peaks from **4pm to midnight** [66]. Incentive schemes also reduced waiting and redirected demand toward low-arrival stations, with users more willing to switch stations than charging times [106].

However, these are mostly **naturalistic or quasi-experimental**, not randomized field trials [66] [68]. Recent evidence also suggests heterogeneous or limited sensitivity in some public-charging settings [44] [67].
Confidence level: Medium-High - real behavior evidence exists, but causal identification is limited [44] [66] [106].

5) Integrated Smart Grid Scheduling Frameworks

The clearest field-wide synthesis is the move toward **integrated forecasting-pricing-scheduling systems** [41] [46]. Joint frameworks report lower peak/load variance and lower user cost than flat or conventional TOU schemes, including **133% higher revenue** and **72.37% lower load variance** versus a uniform rate in one pricing-control system [41]. Tariff-aware scheduling studies report up to **35% peak reduction** and **28% cost savings** [47].

When renewable availability is explicitly included, integrated systems can improve renewable use as well as demand shaping. One dynamic pricing framework aligned low-price periods with high-renewable windows and reported **17.9% higher renewable utilization**, **23.4% average peak-load reduction per station**, and up to **30% user cost savings** versus flat-rate pricing [45]. Another framework reported **25% lower peak demand**, **85% renewable utilization**, **18% lower electricity cost**, and improved voltage stability [48]. Edge/cloud forecasting plus tariff-aware V2G scheduling reduced forecast RMSE by **12.4%** and **18.2%** over baselines while also cutting peak load and cost [46].

Optimization-based energy management further couples forecasts with **MILP, PSO, and two-stage scheduling** for charger assignment and cost control [107] [108]. Broader DER and V2G studies show evidence of **peak shaving, renewable self-consumption, and cost reduction**, including up to **2.8 MWh/day** of EV demand directly met by rooftop solar and renewable fractions up to **50.8%** in hybrid microgrid settings [109] [110].

Taken together, these findings suggest the strongest practical architecture is an integrated control stack that combines **forecasting, tariff setting, scheduling, and DER/V2G coordination** rather than optimizing any one layer in isolation [41] [46] [111].
Confidence level: High - convergence is directly supported across multiple integrated studies [41] [45] [46] [48].

Cross-cutting Synthesis and Practical Takeaways

What appears robust

- **Multi-objective optimization dominates real planning and operations** because EV charging decisions couple service quality, cost, congestion, grid limits, and profitability [8] [29] [55] [112].
- **Contextual data usually help**, especially for siting and spatiotemporal forecasting, but gains depend on data quality and the strength of spatial interaction [2] [20] [50].
- **Queue-aware modeling is valuable** for both operations and pricing because waiting influences station choice and perceived utility [25] [42].
- **Integrated systems outperform decoupled ones** on practical metrics such as peak reduction, renewable use, cost, and revenue [41] [45] [46].

Where the evidence is conflicting

Issue	Evidence for one side	Evidence for the other side	Interpretation
Do deep/graph models always win forecasting?	LSTM and spatiotemporal graph models outperform ARIMA/MLP in some datasets [13] [15].	XGBoost leads RMSE in reproducible benchmarking; spatial layers sometimes add little [14] [20].	No universal winner; horizon, features, and evaluation setup matter [17] [19].

Issue	Evidence for one side	Evidence for the other side	Interpretation
Does TOU pricing strongly shift demand?	Some studies report 50-90% peak-to-off-peak shifting [38].	Others find only 5% peak reduction or limited price suppression in public charging [39] [100] [113].	Pricing works conditionally; waiting, accessibility, and user heterogeneity matter [42] [44].

Confidence level: High - both conflicts are explicitly documented [14] [15] [38] [39].

Conclusion

The existing literature provides a substantial toolkit for EV charging decision support: **location-allocation and evolutionary siting models, temporal and spatiotemporal demand forecasting, queue-aware scheduling and routing, and dynamic pricing/control methods** [10] [13] [29] [41]. The most important trend is not the rise of any single algorithm class, but the convergence toward **integrated, feedback-aware systems** that connect forecasting to siting, assignment, pricing, and smart-grid control [1] [36] [46].

If the goal is real deployment, the evidence supports a practical hierarchy:

1. start with **strong temporal forecasting baselines** and reproducible evaluation [14] [20];
2. add **spatial and queue-aware layers** where congestion spillovers are real [20] [25];
3. optimize using **multi-objective formulations** that include user, operator, and grid KPIs [9] [55] [112];
4. use **adaptive pricing or control** when static tariffs cannot respond to uncertainty [40] [41].

The major remaining weakness is validation: the field still lacks standardized end-to-end benchmarks and enough field-tested evidence to declare a universally best architecture [19] [20] [66].
Overall conclusion confidence: High - the central conclusions are supported across all five thematic areas and reinforced by the meta-analysis [1] [13] [24] [36] [46].

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